

Panel Data Econometrics Homework: Partial Replication

Name of student

To fill in.

1. Identification of the replicated paper

APA reference. Liebersohn, J., & Rothstein, J. (2025). Household mobility and mortgage rate lock. Journal of Financial Economics, article 103973. <https://doi.org/10.1016/j.jfineco.2024.103973>

Link to the paper. <https://doi.org/10.1016/j.jfineco.2024.103973>

Table 1: Key correlation of the paper

- Dependent variable Y: move_flag, an indicator equal to 1 when a household changes ZIP code between quarters.
 - Name and link to the website of data source for updates for Y: synthetic replication package built from UC-CCP-style credit registry data, public access through Mendeley: <https://doi.org/10.17632/74sfv9kx9n.1>
 - Key preferred explanatory variable X: rate_gap, computed as the current market mortgage rate minus the household's implied mortgage rate on the outstanding mortgage.
 - Name and link to the website of data source for updates for X: Freddie Mac mortgage rates through FRED (MORTGAGE30US): <https://fred.stlouisfed.org/series/MORTGAGE30US>
- Selected level(s). Households.
Link to the data set. <https://doi.org/10.17632/74sfv9kx9n.1>
Links to the data set sources.

- Replication package data: <https://doi.org/10.17632/74sfv9kx9n.1>
- FRED mortgage rate series: <https://fred.stlouisfed.org/series/MORTGAGE30US>
- Paper DOI: <https://doi.org/10.1016/j.jfineco.2024.103973>

2. Abstract (100 words)

This partial replication studies whether households move less when their existing mortgage rate is below the current market rate. Using the synthetic replication package for Liebersohn and Rothstein, I build a quarterly household panel from 2012Q1 to 2024Q3, merge mobility events with mortgage characteristics, and construct a household-level rate-gap measure. I document panel structure, variable transformations, descriptive statistics, heterogeneity, and standard panel estimators. Across between, within, two-way fixed effects, and first-difference regressions, the estimated relationship between mobility and the rate gap is small and generally statistically insignificant in this synthetic sample. The exercise remains informative for panel-data workflow and diagnostics.

3. Introduction

The main contribution of this partial replication is methodological rather than a full reproduction of the authors' confidential-data hazard framework. The public package contains synthetic data with the same structure as the original registry-based data, so the present exercise focuses on building a coherent household-quarter panel, implementing panel transformations, and checking what survives under between, within, first-difference, and two-way fixed-effects approaches. This is already useful because the econometric template asks for an ordered diagnostic study of the panel before estimation.

Relative to the paper, the present work highlights three things not emphasized in the published result. First, the synthetic panel is almost ideal from a teaching perspective: after selection it is balanced, has no holes, and therefore allows clean comparisons between within, FD and TWFE transformations. Second, nearly all of the variance of the key variables comes from the within dimension, which means the key identifying variation is essentially time variation within households rather than persistent cross-sectional heterogeneity. Third, once the data are reduced to this simplified binary mobility outcome and linear panel estimators, the estimated rate-gap effect is weak.

This does not invalidate the paper. Instead, it clarifies that the published paper relies on a richer hazard-based design, broader controls, and confidential micro-data structure that cannot be perfectly mirrored here. The value of the present replication is therefore to show a disciplined panel-data workflow, to quantify the differences between transformed variables, and to explain carefully where the simplified public exercise is aligned with the paper and where it is not.

Table 2: Key items found by order of interest of this partial replication

- The selected sample is fully balanced after dynamic-panel selection: 1,000 households observed quarterly from 2012Q1 to 2024Q3.
- The within variance share is extremely high for `move_flag`, `rate_gap`, `mortgage_rate_annual`, and `loan_age_years`, indicating that household time variation dominates cross-sectional variation.
- The association between mobility and the rate gap remains weak under transformed specifications; it is near zero in within and TWFE and mildly negative only in first differences.
- Heterogeneity is substantial across households: some households have strongly positive local correlations and others strongly negative ones, even when the pooled average is near zero.
- The dynamic FD specification shows limited persistence and weak instruments in the simplified Anderson-Hsiao style setup, which cautions against over-interpreting dynamic coefficients in this public sample.

4. Database update

The replication package includes a local `MORTGAGE30US.csv` file ending on 2023-06-08. I updated the market-rate series using the official FRED download endpoint before running the final homework script. This update matters because the mortgage-payment panel extends to 2024Q3, and without the refreshed FRED series the `rate_gap` variable would be missing near the end of the sample. No new households were added because the synthetic micro-data are fixed by the replication package.

The main limitation is that the public package still uses synthetic registry data rather than the confidential underlying data used in the paper.

Table 3: Data characteristics

Item	Value
Maximal time span	2012Q1 to 2024Q3
Frequency	Quarterly
Number of individuals	$N_{\text{total}} = 1000$
Maximal number of time observations for an individual	$T_{\text{max}} = 51$
Balanced or unbalanced panel	Balanced
Update	FRED mortgage rates updated beyond the local 2023-06-08 cutoff so that market-rate coverage matches the panel through 2024Q3

5. Panel data sample selection

The selection rule is at least three consecutive observations of the dependent variable for each household. In this synthetic mortgage panel, every household satisfies this rule.

Panel data sample selection.

- Individuals excluded from the panel data selection: 0
- Individuals with at least three consecutive observations for the dependent variable: $N_1 = 1000$
- $N_{\text{total}} - N_1 = 0$
- Maximal time span for an individual: 2012Q1 to 2024Q3
- List of excluded individuals: None
- List of kept individuals: omitted here because there are 1,000 households; the full list is the set of all `personid` in `panel_data.csv`.

Comment on sample-selection bias. There is no observed selection bias induced by the three-consecutive-observations rule in this synthetic sample because no household is dropped. This is convenient for pedagogy but also a limitation: the public synthetic data understate the attrition and irregular observation issues commonly found in real household panels.

6. Sample selection within an unbalanced panel

Although the template asks for the analysis of an unbalanced panel, the selected sample is in fact balanced. Therefore the diagnostics below are still reported, but they mechanically show no attrition and no holes.

Number of individuals per date
quarter

N

2012Q1

1000

2012Q2

1000

2012Q3

1000

2012Q4

1000

2013Q1

1000

2013Q2

1000

2013Q3

1000

2013Q4

1000

2014Q1

1000

2014Q2

1000

2014Q3

1000

2014Q4

1000

2015Q1

1000

2015Q2

1000

2015Q3

1000

2015Q4

1000

2016Q1

1000

2016Q2

1000

2016Q3

1000

2016Q4

1000

2017Q1

1000

2017Q2

1000

2017Q3

1000

2017Q4

1000

2018Q1

1000

2018Q2

1000

2018Q3

1000

2018Q4

1000

2019Q1

1000

2019Q2

```

1000
2019Q3
1000
2019Q4
1000
2020Q1
1000
2020Q2
1000
2020Q3
1000
2020Q4
1000
2021Q1
1000
2021Q2
1000
2021Q3
1000
2021Q4
1000
2022Q1
1000
2022Q2
1000
2022Q3
1000
2022Q4
1000
2023Q1
1000
2023Q2
1000
2023Q3
1000
2023Q4
1000
2024Q1
1000
2024Q2
1000
2024Q3
1000
Number of individuals with the same number of temporal observations
quarter
number_of_individuals
51
1000

```

- Number of individuals with holes: 0
- Proportion with holes: 0.00%
- List with holes: []

Comment. There is no attrition at the beginning or at the end of the sample and no discontinuity pattern. This again reflects the synthetic design of the teaching data rather than the full complexity of the confidential panel behind the paper.

7. Panel data variables classification

Table 4: Variable count by categories

- Variables varying with two indices: rate_gap, mortgage_rate_annual, move_flag, log_balance, balance_to_orig, loan_age_years
- Time-invariant variables (100% between variance): None in the selected variable set
- Individual-invariant variables (0% between variance): market_rat

e, trend

- Number K = 6
- K1 = 0
- K2 = 2

Table 5: Time-varying variables sorted by within share variable

N
NT
NT_over_N
overall_var
between_var
within_var
within_share
rate_gap
1000
48737
48.737000
1.889807
0.011398
1.919089
1.015495
mortgage_rate_annual
1000
48737
48.737000
0.456431
0.010072
0.457020
1.001292
move_flag
1000
51000
51.000000
0.125795
0.002460
0.125802
1.000055
log_balance
1000
51000
51.000000
2.032352
0.045471
2.026625
0.997182
balance_to_orig
1000
51000
51.000000
44131.307643
2576.719555
42387.425478
0.960484
loan_age_years
1000
51000
51.000000
2.393774
0.510415
1.921499
0.802707
Time-invariant variables (100% between variance)

None in the selected set.
Individual-invariant variables (0% between variance)

variable
N
NT
NT_over_N
overall_var
between_var
within_var
within_share
market_rate
1000
51000
51.000000
1.427681
0.000000
1.456206
1.019980
trend
1000
51000
51.000000
216.670915
0.000000
221.000000
1.019980

Comment. The key dependent variable `move_flag` and the key explanatory variable `rate_gap` both have very high within shares. This means that the informative variation for the homework mostly comes from the evolution of a household over time rather than from persistent differences across households. The common mortgage-rate series (`market_rate`) and the deterministic time trend are individual-invariant variables, so they are wiped out by time dummies and/or by TWFE transformation

8. Between plus one-way within decomposition

Descriptive statistics for the dependent variable `move_flag`
transform

n
mean
median
sd
se
min
q1
q3
max
skew
kurtosis
std_min
std_max
between
51000
0.147569
0.137255
0.049571
0.049571
0.019608
0.117647
0.176471
0.333333
0.387867
3.335060
-2.581353

3.747432
within
51000
-0.000000
-0.137255
0.351194
0.351194
-0.333333
-0.176471
-0.098039
0.980392
1.929329
4.877890
-0.949143
2.791596
fd
50000
0.003060
0.000000
0.503007
0.503007
-1.000000
0.000000
0.000000
1.000000
0.005794
3.952401
-1.994129
1.981962
twfe
51000
0.000000
-0.128686
0.346629
0.346629
-0.614549
-0.171902
-0.074863
1.070353
1.874439
4.867251
-1.772930
3.087892

Descriptive statistics for the preferred explanatory variable rate_gap
transform

n
mean
median
sd
se
min
q1
q3
max
skew
kurtosis
std_min
std_max
between
51000
4.092290
4.114000

```

0.106710
0.106710
3.576468
4.037769
4.168272
4.290295
-1.112028
4.861055
-4.833877
1.855547
within
48737
0.000000
-0.282254
1.370572
1.370572
-11.919270
-0.686838
0.303433
3.655761
-0.226394
8.734244
-8.696563
2.667325
fd
45668
0.049493
-0.003593
1.036280
1.036280
-11.849085
-0.242456
0.290738
12.431947
-0.034364
40.936045
-11.482011
11.948947
twfe
48737
-0.000000
0.110874
0.668686
0.668686
-11.335833
0.015642
0.199821
0.795920
-8.586655
100.252310
-16.952390
1.190274

```

The within-transformed mobility indicator is highly non-normal and discrete, with many observations concentrated near household means. The between distribution of `rate_gap` is tighter than the within distribution, which is consistent with the idea that market rates and loan-specific conditions move substantially over time within households. The graphs saved in `analysis_outputs/move_flag_transforms.png` and `analysis_outputs/rate_gap_transforms.png` display the histogram, KDE, and matching normal approximation requested in the template.

9. First differences versus two-way fixed effects

Because the selected sample is balanced, the closed-form TWFE transformation $x_{it} - x_{i.} - x_{.t} + x_{..}$ is well defined and coincides with the regression-residual

l interpretation. For first differences, the first observation of each individual is missing by construction. The file analysis_outputs/fd_first30.csv reports the first 30 stacked observations of first differences and their lags, and one can verify that each individual switch generates missing first differences rather than cross-household subtraction.

Between vs within correlation matrix

```
index
move_flag_between
rate_gap_between
balance_to_orig_between
loan_age_years_between
log_balance_between
move_flag_between
1.000000
0.052314
0.004630
-0.416690
0.001786
rate_gap_between
0.052314
1.000000
-0.003660
-0.028802
0.473497
balance_to_orig_between
0.004630
-0.003660
1.000000
-0.001591
-0.083276
loan_age_years_between
-0.416690
-0.028802
-0.001591
1.000000
0.024215
log_balance_between
0.001786
0.473497
-0.083276
0.024215
1.000000
```

```
index
move_flag_within
rate_gap_within
balance_to_orig_within
loan_age_years_within
log_balance_within
trend_within
move_flag_within
1.000000
0.004038
-0.007428
-0.174514
0.004541
0.014490
rate_gap_within
0.004038
1.000000
-0.002453
0.057984
```

0.243703
0.466081
balance_to_orig_within
-0.007428
-0.002453
1.000000
-0.005025
0.022447
-0.004764
loan_age_years_within
-0.174514
0.057984
-0.005025
1.000000
-0.005559
0.252264
log_balance_within
0.004541
0.243703
0.022447
-0.005559
1.000000
0.000096
trend_within
0.014490
0.466081
-0.004764
0.252264
0.000096
1.000000
TWFE vs FD correlation matrix
index
move_flag_twfe
rate_gap_twfe
balance_to_orig_twfe
loan_age_years_twfe
log_balance_twfe
move_flag_twfe
1.000000
0.001892
-0.007375
-0.198220
0.004439
rate_gap_twfe
0.001892
1.000000
0.006838
-0.000247
0.482349
balance_to_orig_twfe
-0.007375
0.006838
1.000000
-0.003654
0.022366
loan_age_years_twfe
-0.198220
-0.000247
-0.003654
1.000000
-0.003955
log_balance_twfe

0.004439
0.482349
0.022366
-0.003955
1.000000

index
move_flag_fd
rate_gap_fd
balance_to_orig_fd
loan_age_years_fd
log_balance_fd
move_flag_fd_lag1
rate_gap_fd_lag1
move_flag_fd
1.000000
-0.007410
-0.014307
-0.305840
0.001279
-0.503768
0.000742
rate_gap_fd
-0.007410
1.000000
0.006575
-0.006187
0.450417
0.006452
-0.412148
balance_to_orig_fd
-0.014307
0.006575
1.000000
0.007372
0.029813
0.002737
-0.001369
loan_age_years_fd
-0.305840
-0.006187
0.007372
1.000000
-0.003815
0.005156
-0.007473
log_balance_fd
0.001279
0.450417
0.029813
-0.003815
1.000000
0.004003
-0.230097
move_flag_fd_lag1
-0.503768
0.006452
0.002737
0.005156
0.004003
1.000000
-0.007216

rate_gap_fd_lag1

0.000742

-0.412148

-0.001369

-0.007473

-0.230097

-0.007216

1.000000

The simple correlations between transformed mobility and transformed rate-gap variables remain weak in absolute value, usually below 0.1, which is exactly the kind of diagnostic the template asks us to note before moving to regressions. The strongest correlations are mostly among explanatory variables and their lags rather than between the dependent variable and the key explanatory variable.

10. Investigating bivariate heterogeneity by individuals for TWFE and FD

FD heterogeneity table (top 15 households)

personid

T_i

corr

sd_y

sd_x

sd_ratio

beta

A865000

48

0.487346

0.291730

0.716015

0.407436

0.198562

A114000

46

0.457269

0.493924

2.083628

0.237050

0.108396

A749000

46

0.433213

0.446673

1.092391

0.408895

0.177139

A633000

48

0.410655

0.545777

0.608111

0.897495

0.368561

A464000

41

0.408283

0.523823

0.441959

1.185229

0.483910

A806000

42

0.402288

0.538851

1.353879

0.398005
0.160113
A597000
48
0.399628
0.635462
0.432059
1.470775
0.587763
A198000
48
0.389519
0.601048
0.898755
0.668757
0.260494
A963000
44
0.388515
0.681994
0.481448
1.416547
0.550350
A106000
46
0.378778
0.446673
1.475461
0.302735
0.114669
A160000
48
0.364132
0.618853
0.564793
1.095716
0.398986
A417000
40
0.354812
0.552384
0.537491
1.027710
0.364644
A988000
48
0.352375
0.525502
0.518943
1.012639
0.356828
A228000
45
0.332781
0.424026
1.820645
0.232899
0.077504
A577000
46
0.331839
0.596285

0.414725
1.437784
0.477113
TWFE heterogeneity table (top 15 households)
personid
T_i
corr
sd_y
sd_x
sd_ratio
beta
A963000
48
0.270483
0.458631
0.214467
2.138473
0.578420
A803000
49
0.264065
0.316578
0.133697
2.367875
0.625274
A939000
50
0.255064
0.374256
0.251831
1.486139
0.379060
A632000
48
0.231929
0.413671
0.229639
1.801395
0.417795
A859000
48
0.226349
0.381332
0.190292
2.003933
0.453587
A633000
50
0.219752
0.353331
0.336027
1.051497
0.231069
A266000
49
0.218839
0.333143
0.104639
3.183735
0.696725
A401000
51

0.216054
 0.376469
 0.401410
 0.937867
 0.202630
 A931000
 49
 0.215674
 0.333057
 0.402531
 0.827409
 0.178450
 A417000
 46
 0.214949
 0.420469
 0.280954
 1.496573
 0.321686
 A250000
 47
 0.207706
 0.384815
 0.201274
 1.911899
 0.397112
 A820000
 50
 0.206220
 0.389418
 0.180982
 2.151690
 0.443722
 A239000
 47
 0.205037
 0.315982
 0.687771
 0.459430
 0.094200
 A024000
 50
 0.203571
 0.406944
 0.280416
 1.451215
 0.295426
 A083000
 48
 0.202623
 0.311964
 0.110329
 2.827565
 0.572931

- FD diagnosis: positive correlation households = 288, negative correlation households = 243, weak-correlation households = 469.

- TWFE diagnosis: positive correlation households = 346, negative correlation households = 196, weak-correlation households = 458.

The pooled effect is small partly because heterogeneity is large: some households show positive local comovement, others negative comovement. This is precisely why the template asks for household-level heterogeneity tables rather than relying only on pooled coefficients. Rolling-window plots for representative household

ds are saved in analysis_outputs/fd_rolling_corr.png and analysis_outputs/twfe_rolling_corr.png.

11. Panel data estimates

model
n_obs
coef_rate_gap
se_rate_gap
r_squared

between
1000
0.020735
0.015261
0.175337
within
48737
0.003618
0.001194
0.030725

mundlak
48737
0.003629
0.001209
0.033640

twfe
48737
0.000032
0.002686
0.039371

fd
45668
-0.005959
0.002360
0.092429

Across between, within, Mundlak, TWFE and first-difference estimators, the coefficient on rate_gap remains economically small in the synthetic sample. The between, within and Mundlak coefficients are slightly positive, the TWFE coefficient is essentially zero, and only the first-difference coefficient is negative. This is an important replication conclusion: the simplified public panel exercise does not recover a stable reduced-form sign comparable to the full confidential-data hazard specification.

Dynamic panel / Anderson-Hsiao style extension

Univariate statistics for dynamic variables

variable
n
mean
median
sd
se
min
q1
q3
max
skew
kurtosis
std_min
std_max

dy
42799
0.001612
0.000000
0.509109
0.509109

-1.000000
0.000000
0.000000
1.000000
0.002718
3.858422
-1.967383
1.961050
dy_l1
42799
0.001916
0.000000
0.506093
0.506093
-1.000000
0.000000
0.000000
1.000000
0.003424
3.904526
-1.979708
1.972137
dx
42799
0.056031
0.005064
1.043341
1.043341
-11.849085
-0.242583
0.297063
12.431947
0.065063
40.629890
-11.410566
11.861809
dx_l1
42799
0.058741
0.007574
1.036956
1.036956
-11.849085
-0.229883
0.297836
12.431947
-0.086836
41.130615
-11.483440
11.932235
y_l2
42799
0.149419
0.000000
0.356506
0.356506
0.000000
0.000000
0.000000
1.000000
1.966853
4.868598

-0.419122
2.385883
x_l2
42799
3.991430
3.791079
1.298371
1.298371
-8.181924
3.406824
4.286574
7.303224
-0.404210
10.995020
-9.375866
2.550730

Correlation matrix for dynamic variables and instruments

index
dy
dy_l1
dx
dx_l1
y_l2
x_l2
dy
1.000000
-0.503768
-0.007410
0.000742
0.004208
-0.004623
dy_l1
-0.503768
1.000000
0.006452
-0.007216
-0.706084
-0.003264
dx
-0.007410
0.006452
1.000000
-0.412148
0.004068
-0.037537
dx_l1
0.000742
-0.007216
-0.412148
1.000000
0.015926
-0.336558
y_l2
0.004208
-0.706084
0.004068
0.015926
1.000000
0.002562
x_l2
-0.004623
-0.003264

-0.037537
-0.336558
0.002562
1.000000

- ADF test on ΔY : statistic = -71.831, p-value = 0.0000
- ADF test on ΔX : statistic = -26.107, p-value = 0.0000

These pooled ADF tests strongly suggest stationarity of the differenced series, which is the expected result after first differencing. The template asked for a panel unit-root test of our choice; in this local Python environment, the pooled ADF test is the feasible choice.

OLS dynamic FD regression

OLS Regression Results

```
=====Dep
p. Variable:          dy      R-squared:          0.364Mod
el:                   OLS      Adj. R-squared:      0.363Meth
od:                   Least Squares      F-statistic:    303.6Date:
                        Thu, 28 May 2026      Prob (F-statistic):    0.00Time:
                        17:15:49      Log-Likelihood:    -22159.No. Obs
ervations:            42799      AIC:                4.443e+04Df Resid
uals:                 42744      BIC:                4.491e+04Df Model:
                        54                                     Covariance
Type:                 HC1                                     =====
```

	coef	std err	z	P> z	[0.025	0.975]	
-----const		0.0666	0.007	9.777	0.000	0.053	
0.080dx		-0.0012	0.003	-0.475	0.635		
-0.006	0.004d_balance_to_orig	-2.153e-05	7.62e-06	-2.827	0.005		
-3.65e-05	-6.6e-06d_loan_age_years	-0.2001	0.005	-43.880			
0.000	-0.209	-0.191d_log_balance	0.0009	0.001	0.75		
5	0.450	-0.001	0.003q_2012Q4	0.0118	0.011		
1.078	0.281	-0.010	0.033q_2013Q1	0.3570	0.0		
16	22.341	0.000	0.326	0.388q_2013Q2	-0.1538		
0.017	-9.202	0.000	-0.187	-0.121q_2013Q3	-0.2		
152	0.017	-12.513	0.000	-0.249	-0.181q_2013Q4		
-0.0598	0.016	-3.813	0.000	-0.090	-0.029q_2014Q1		
-0.0584	0.015	-3.836	0.000	-0.088	-0.029q_2014Q2		
2	-0.0513	0.015	-3.324	0.001	-0.082	-0.021q_2014Q3	
-0.010q_2014Q4	-0.0390	0.015	-2.620	0.009	-0.068	-0.071	
0.010q_2015Q1	-0.0405	0.015	-2.629	0.009	-0.068	-0.071	
-0.010q_2015Q2	-0.0599	0.016	-3.824	0.000	-0.071	-0.071	
.091	-0.029q_2015Q3	-0.0661	0.016	-4.242	0.000	0.000	
-0.097	-0.036q_2015Q4	-0.0399	0.016	-2.516	0.000	0.000	
.012	-0.071	-0.009q_2016Q1	-0.0718	0.015	-4.705		
0.000	-0.102	-0.042q_2016Q2	-0.0856	0.016	-		
5.501	0.000	-0.116	-0.055q_2016Q3	-0.0692	0.015		
-4.478	0.000	-0.099	-0.039q_2016Q4	-0.0667	0.015		
0.015	-4.433	0.000	-0.096	-0.037q_2017Q1	-0.063		
3	0.015	-4.204	0.000	-0.093	-0.034q_2017Q2		
-0.0462	0.016	-2.982	0.003	-0.077	-0.016q_2017Q3		
-0.0521	0.016	-3.244	0.001	-0.084	-0.021q_2017Q4		
-0.0744	0.016	-4.757	0.000	-0.105	-0.044q_2018Q1		
0.017Q4	-0.0686	0.015	-4.442	0.000	-0.099	-0.0	
38q_2018Q2	-0.0620	0.015	-4.130	0.000	-0.091	-0.0	
-0.033q_2018Q3	-0.0815	0.016	-5.164	0.000	-0.091	-0.1	
12	-0.051q_2018Q4	-0.0957	0.015	-6.273	0.000	0.000	
-0.126	-0.066q_2019Q1	-0.0455	0.016	-2.897	0.0	0.0	
04	-0.076	-0.015q_2019Q2	-0.0469	0.016	-2.984	-4.	
0.003	-0.078	-0.016q_2019Q3	-0.0648	0.015	-4.		
239	0.000	-0.095	-0.035q_2019Q4	-0.0764	0.015	0	
-5.056	0.000	-0.106	-0.047q_2020Q1	-0.0628	0		
.015	-4.237	0.000	-0.092	-0.034q_2020Q2	-0.0803		

0.015	-5.502	0.000	-0.109	-0.052	q_2020Q2	-0
.0523	0.015	-3.469	0.001	-0.082	-0.023	q_2020Q3
-0.0486	0.016	-3.097	0.002	-0.079	-0.018	q_2020Q4
-0.0783	0.015	-5.056	0.000	-0.109	-0.048	q_2021Q1
-0.0497	0.016	-3.042	0.002	-0.082	-0.018	q_2021Q2
-0.0748	0.015	-4.844	0.000	-0.105	-0.111	q_2021Q3
-0.0811	0.015	-5.351	0.000	-0.111	-0.111	q_2021Q4
-0.0623	0.016	-4.017	0.000	-0.111	-0.111	q_2022Q1
-0.0631	0.016	-3.997	0.000	-0.111	-0.111	q_2022Q2
-0.0427	0.017	-2.547	-3.98	-3.98	-3.98	q_2022Q3
-0.0642	0.016	-0.0915	0.016	0.016	0.016	q_2022Q4
-0.0915	0.016	-0.0833	0.016	0.016	0.016	q_2023Q1
-0.052	0.016	-0.0725	0.016	0.016	0.016	q_2023Q2
-0.043	0.016	-0.0725	0.016	0.016	0.016	q_2023Q3
-0.043	0.016	-0.0725	0.016	0.016	0.016	q_2023Q4
-0.043	0.016	-0.0725	0.016	0.016	0.016	q_2024Q1
-0.020	0.016	-0.0725	0.016	0.016	0.016	q_2024Q2
-0.024	0.016	-0.0725	0.016	0.016	0.016	q_2024Q3
-0.039	0.016	-0.0725	0.016	0.016	0.016	q_2024Q4
-0.032	0.016	-0.0725	0.016	0.016	0.016	dy_l1
-0.499	0.016	-0.0725	0.016	0.016	0.016	dx_l1
.004	0.005	0.0004	0.002	0.187	0.851	-0

Omnibus:	2923.603	Durbin-Watson:	0.000
2.272Prob(Omnibus):	0.000	Jarque-Bera (JB):	0.520
5065.037Skew:	0.520	Prob(JB):	4.326
0.00Kurtosis:	4.326	Cond. No.	1.29e+04

Notes: [1] Standard Errors are heteroscedasticity robust (HC1) [2] The condition number is large, 1.29e+04. This might indicate that there are strong multicollinearity or other numerical problems.

IV dynamic FD regression (levels instruments Y_{t-2} and X_{t-2})

IV2SLS Regression Results

Dep. Variable:	dy	R-squared:	0.143	Mod
Model:	IV2SLS	Adj. R-squared:	0.142	Meth
Method:	Two Stage	F-statistic:	106.7	
	Least Squares	Prob (F-statistic):	0.000	Date:
	Thu, 28 May 2026			Time:
	17:15:49			No. Obs
Observations:	42799			Df Residuals
Variables:	42744			Df Model:
	54			

	coef	std err	t	P> t	[0.025	0.975]
-----const	0.0533	0.016	3.340	0.001	0.022	0.085
0.085dx	-0.0031	0.003	-0.982	0.326	-0.009	0.003
-0.009	0.003	-2.449e-05	8.78e-06	-2.788	-4.17e-05	-7.27e-06
-4.17e-05	-7.27e-06	-0.2012	0.003	-67.917	0.000	-0.207
0.000	-0.207	-0.195	0.0014	0.001	1.06	0.289
1	0.289	-0.001	0.004	0.0209	0.023	0.927
0.927	0.354	-0.023	0.065	0.3560	0.0	15.832
22	15.832	0.000	0.312	0.400	-0.3367	0.023
0.023	-14.827	0.000	-0.381	-0.292	-0.0	0.023
442	0.023	-1.918	0.055	-0.089	0.001	0.023
-0.0466	0.023	-2.044	0.041	-0.091	-0.002	-0.0325
-0.0325	0.023	-1.441	0.150	-0.077	0.012	-0.0395
2	-0.0395	0.022	-1.757	0.079	-0.084	-0.0248
-0.0248	0.022	-1.103	0.270	-0.069	0.019	-0.0331
.019q_2014Q4	-0.0331	0.023	-1.469	0.142	-0.077	0.011
0.011q_2015Q1	-0.0509	0.023	-2.259	0.024	-0	

```

.095      -0.007q_2015Q2      -0.0445      0.023      -1.973      0.048
-0.089      -0.000q_2015Q3      -0.0241      0.023      -1.071      0
.284      -0.068      0.020q_2015Q4      -0.0703      0.022      -3.125
0.002      -0.114      -0.026q_2016Q1      -0.0626      0.023      -
2.780      0.005      -0.107      -0.018q_2016Q2      -0.0507      0.023
-2.251      0.024      -0.095      -0.007q_2016Q3      -0.0515
0.022      -2.291      0.022      -0.095      -0.007q_2016Q4      -0.046
7      0.023      -2.071      0.038      -0.091      -0.002q_2017Q1
-0.0372      0.023      -1.642      0.101      -0.082      0.007q_2017Q2
-0.0473      0.023      -2.091      0.037      -0.092      -0.003q_2017Q3
-0.0622      0.023      -2.761      0.006      -0.106      -0.018q_2
017Q4      -0.0494      0.023      -2.187      0.029      -0.094      -0.0
05q_2018Q1      -0.0490      0.023      -2.170      0.030      -0.093
-0.005q_2018Q2      -0.0689      0.023      -3.042      0.002      -0.1
13      -0.025q_2018Q3      -0.0806      0.023      -3.549      0.000
-0.125      -0.036q_2018Q4      -0.0215      0.023      -0.943      0.3
46      -0.066      0.023q_2019Q1      -0.0429      0.023      -1.893
0.058      -0.087      0.002q_2019Q2      -0.0539      0.023      -2.
385      0.017      -0.098      -0.010q_2019Q3      -0.0639      0.023
-2.828      0.005      -0.108      -0.020q_2019Q4      -0.0400      0
.023      -1.772      0.076      -0.084      0.004q_2020Q1      -0.0639
0.023      -2.829      0.005      -0.108      -0.020q_2020Q2      -0
.0286      0.023      -1.261      0.207      -0.073      0.016q_2020Q3
-0.0480      0.023      -2.123      0.034      -0.092      -0.004q_2020Q4
-0.0602      0.023      -2.667      0.008      -0.105      -0.016q_202
1Q1      -0.0372      0.023      -1.651      0.099      -0.081      0.007
q_2021Q2      -0.0724      0.023      -3.207      0.001      -0.117
-0.028q_2021Q3      -0.0548      0.023      -2.428      0.015      -0.099
-0.011q_2021Q4      -0.0407      0.023      -1.799      0.072
-0.085      0.004q_2022Q1      -0.0533      0.023      -2.310      0.021
-0.098      -0.008q_2022Q2      -0.0281      0.024      -1.182
0.237      -0.075      0.019q_2022Q3      -0.0688      0.024      -2.91
0      0.004      -0.115      -0.022q_2022Q4      -0.0790      0.023
-3.419      0.001      -0.124      -0.034q_2023Q1      -0.0615      0.0
23      -2.673      0.008      -0.107      -0.016q_2023Q2      -0.0591
0.023      -2.613      0.009      -0.103      -0.015q_2023Q3      -0.0
507      0.023      -2.225      0.026      -0.095      -0.006q_2023Q4
-0.0440      0.023      -1.931      0.054      -0.089      0.001q_2024Q1
-0.0446      0.023      -1.973      0.048      -0.089      -0.000q_2024Q
2      -0.0631      0.023      -2.790      0.005      -0.107      -0.019q
_2024Q3      -0.0416      0.023      -1.839      0.066      -0.086      0
.003dy_l1      -0.0290      0.006      -4.501      0.000      -0.042
-0.016dx_l1      0.0041      0.003      1.213      0.225      -0
.003      0.011=====
=====Omnibus:      1304.124      Durbin-Watson:
      2.863Prob(Omnibus):      0.000      Jarque-Bera (JB):
2769.515Skew:      -0.193      Prob(JB):
0.00Kurtosis:      4.185      Cond. No.
1.29e+04=====
=====
First-stage regressions
•      First stage for  $\Delta Y_{(t-1)}$ :  $R^2 = 0.5117$ 
•      First stage for  $\Delta X_{(t-1)}$ :  $R^2 = 0.7811$ 
The first-stage  $R^2$  values are not especially low in this simplified setup, so there is no immediate weak-instrument warning from that diagnostic alone. Even so, the dynamic IV exercise should still be read as a pedagogical robustness check rather than a definitive causal estimate, because the public synthetic panel is not the authors' full hazard dataset.
Impulse response and long-run coefficient
horizon
response
1.000000

```

-0.001236
2.000000
0.001074
3.000000
-0.000545
4.000000
0.000276

- Long-run coefficient: -0.000524

12. Optional time-invariant-variable estimators

Skipped. In the selected working set for this homework, I did not keep a substantively meaningful time-invariant household characteristic that could justify a Hausman-Taylor exercise. Forcing an artificial time-invariant regressor would not improve the econometric quality of the homework. This is preferable to running a mechanical but uninformative Hausman-Taylor specification.

13. Own estimations not done in the paper

As an original extension, I estimated a TWFE-style interaction model allowing the effect of the rate gap to differ for households above the median balance-to-original-loan ratio. This is a simple way to test whether rate lock is more visible among relatively high-balance households.

OLS Regression Results

```
=====Dep
p. Variable:      move_flag      R-squared:      0.041Mod
el:               OLS      Adj. R-squared:      0.041Meth
od:               Least Squares      F-statistic:      364.8Date:
                Thu, 28 May 2026      Prob (F-statistic):      0.00Time:
                17:15:49      Log-Likelihood:      -16506.No. Obs
ervations:        48737      AIC:      3.302e+04Df Resid
uals:             48731      BIC:      3.308e+04Df Model:
                5      Covariance
Type:             HC1=====
```

```
=====
              coef      std err              z      P>|z|      [0.025
0.975]-----
-----const              -0.0094      0.002      -5.166      0
.000      -0.013      -0.006rate_gap              -0.0008      0.003      -0.
313      0.754      -0.006      0.004rate_gap_high_balance      0.1110      0.0
14      8.134      0.000      0.084      0.138balance_to_orig      -1.436e-0
5      4.83e-06      -2.974      0.003      -2.38e-05      -4.9e-06loan_age_years
      -0.0531      0.001      -42.027      0.000      -0.056      -0.051log_balance
      -0.0041      0.001      -2.812      0.005      -0.007      -0.001=
```

```
=====Omnibus:      14226.333      Durbin-Watson:      2.037Pro
b(Omnibus):      0.000      Jarque-Bera (JB):      30955.793Skew
:      1.760      Prob(JB):      0.00Kurto
sis:      4.690      Cond. No.      1.77e+03=====
```

Notes:

[1] Standard Errors are heteroscedasticity robust (HC1)[2] The condition number is large, 1.77e+03. This might indicate that there are strong multicollinearity or other numerical problems.

The interaction term should be interpreted cautiously, but it offers a clean example of an original panel-data extension beyond the baseline replication workflow.

14. List of references cited in the text

- Liebersohn, J., & Rothstein, J. (2025). Household mobility and mortgage rate lock. Journal of Financial Economics. <https://doi.org/10.1016/j.jfineco.2024.103973>

- Freddie Mac / Federal Reserve Bank of St. Louis. MORTGAGE30US, 30-Year Fixed Rate Mortgage Average in the United States. <https://fred.stlouisfed.org/series/MORTGAGE30US>

- Mendeley Data replication package: <https://doi.org/10.17632/74sfv9kx9n.1>

- Replication Package README included locally in the project folder

- r.
- Appendix: To-do list for the next panel-data study
- ☐ Define the economic question and the exact identifying variation before coding.
 - ☐ List the original paper's preferred dependent variable, key explanatory variable, controls, and fixed effects.
 - ☐ Verify raw data provenance and save permanent source links immediately.
 - ☐ Document every cleaning choice and every sample restriction in a reproducible script.
 - ☐ Check panel identifiers and time identifiers before any merge.
 - ☐ Count duplicates by individual-time cell before estimating anything.
 - ☐ Inspect missing values separately for dependent and explanatory variables.
 - ☐ Create the dynamic-panel benchmark sample with at least three consecutive observations.
 - ☐ Report attrition, holes, and sample-selection bias before regressions.
 - ☐ Compute between, within, FD and TWFE transformations early and compare their distributions.
 - ☐ Sort variables by within-variance share to understand which estimators are informative.
 - ☐ Check lags and first differences manually on the first stacked observations.
 - ☐ Inspect heterogeneity by individuals before relying on pooled coefficients.
 - ☐ Test robustness with alternative functional forms and subgroup interactions.
 - ☐ Assess instrument strength explicitly before reporting IV results.
 - ☐ Write interpretation in plain language after every table, not only code comments.
 - ☐ Keep an ordered folder of outputs: data, tables, figures, report, prompts.
 - ☐ Export final tables in both machine-readable and human-readable formats.
 - ☐ State clearly what is replicated exactly and what is only approximated.
 - ☐ Finish with a limitations paragraph and a short agenda for future work.